



Short communication

High-performance solid polymer electrolytes for lithium batteries operational at ambient temperature

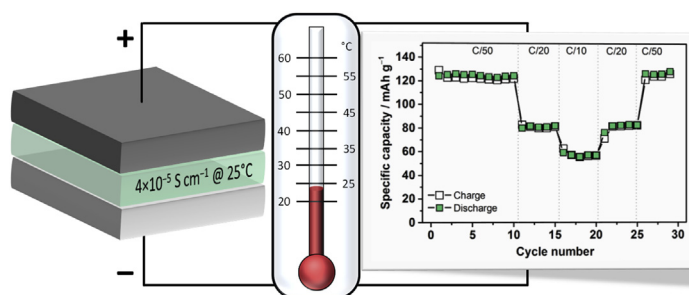
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HIGHLIGHTS

- Polymer electrolytes were prepared based on poly(ϵ -caprolactone).
- Optimal performance was attained by addition of 20 mol% carbonate repeating units.
- High total ionic conductivity; $4.1 \times 10^{-5} \text{ S cm}^{-1}$ at 25°C , $1.1 \times 10^{-4} \text{ S cm}^{-1}$ at 40°C .
- High Li^+ transference numbers; 0.62 at 40°C , 0.66 at 60°C .
- Li polymer half-cells were cycled with high stability at room temperature.

GRAPHICAL ABSTRACT



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ABSTRACT

Incorporation of carbonate repeating units in a poly(ϵ -caprolactone) (PCL) backbone used as a host material in solid polymer electrolytes is found to not only suppress crystallinity in the polyester material, but also give higher ionic conductivity in a wide temperature range exceeding the melting point of PCL crystallites. Combined with high cation transference numbers, this electrolyte material has sufficient lithium transport properties to be used in battery cells that are operational at temperatures down to below 23°C , thus clearly demonstrating the potential of using non-polyether electrolytes in high-performance all-solid lithium polymer batteries.

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1. Introduction

One major challenge in the development of Li batteries is finding a suitable electrolyte [1]. The low-molecular-weight organics currently used as electrolyte solvents are inherently unstable at the electrode/electrolyte interfaces, leading to irreversible loss of capacity during the first cycle due to the formation of a solid

electrolyte interphase (SEI) layer as well as the risk of continuous further decomposition of solvent and salt during the operational lifetime of the battery [2,3]. The volatility and flammability of these solvents are also sources of great concern, as runaway reactions in such batteries may have devastating consequences [4]; notable examples being the recent Boeing 787 Dreamliner battery incidents [5]. One possible way to mitigate these problems is to replace the liquids with solid-phase electrolytes with concomitant improvements in stability and safety while also providing enhanced flexibility in cell design [6,7].

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Ever since first suggested by Armand et al. [8,9], solid (solvent-free) polymer electrolytes (SPEs) have been intensively researched for potential use in Li battery applications. Compared to other solid electrolytes, SPEs are potentially both low-cost and easily processable [1,3,6]. Furthermore, they offer all the versatility and flexibility of polymer chemistry, allowing for facile synthesis of tailored materials with specific functionalities for specific applications, ranging from self-assembling crosslinkable oligomers for 3D-microbatteries [10] to electrolytes with tailored mechanical properties for structural batteries [11].

The vast majority of Li⁺-conducting SPEs are based on polyethers, and in particular the oxyethylene motif; the most prominent example being poly(ethylene oxide) (PEO). Few alternatives exist that are not subpar to this standard material [6] and most electrolytes typically fall short of either the demand for high ionic conductivity or sufficient mechanical properties [1,3,11,12]. In the case of PEO, this is exacerbated by semicrystallinity at low to moderate temperatures, which severely restricts ion transport in the material [13]. Other issues that have been highlighted include compatibility with the highly porous electrodes and good wetting of the active materials [3]. These challenges have so far limited the practical use of SPEs and despite the wealth of literature on solid polymer electrolytes, there is a notable lack of cycling data.

We and other authors have recently started investigating polycarbonates as alternative host materials for SPEs [14,15]. Although polycarbonate electrolytes have shown good stability and cycling efficiency in Li half-cells at elevated temperature, the low ionic conductivity currently limits the use of these materials to low-power and/or high-temperature applications [14,16]. We were recently able to demonstrate increased ionic conductivity and better cycling performance for electrolytes based on poly(trimethylene carbonate) (PTMC) on incorporation of ϵ -caprolactone repeating units in the polymer chains (from 10 to 40 mol%), attributed to increased chain flexibility [17]. Based on these results, it could be anticipated that optimal performance would be obtained in electrolytes based on the poly(ϵ -caprolactone) (PCL) homopolymer [18–20]. However, while showing decent ionic conductivity at elevated temperatures, much like PEO, polymer electrolytes based on PCL are semicrystalline at room temperature, thereby severely limiting their ionic conductivity. In this short communication, we demonstrate how crystallinity can be suppressed in PCL-based polymer electrolytes by copolymerization with trimethylene carbonate (TMC), thereby improving the ionic conductivity in a broad temperature range to the point where use in batteries functional at room temperature becomes feasible.

2. Experimental

2.1. Materials

ϵ -Caprolactone (CL; Aldrich) was distilled under reduced pressure over CaH₂ and stored in a glovebox under argon. Trimethylene carbonate (TMC; Boehringer Ingelheim) was handled and stored in a glove box under argon and used as received. Copolymers of CL and TMC were synthesized through bulk ring-opening polymerization according to the same procedure as previously published for the homopolymerization of TMC [14]. Lithium bis(trifluoromethanesulfonimide) (LiTFSI; Purolite, Ferro Corporation) was dried in vacuum at 120 °C for 24 h before use. All other chemicals were obtained from commercial sources and used as received.

2.2. Polymer electrolyte preparation

Electrolyte films were prepared by casting from solutions in anhydrous acetonitrile (0.5 g polymer in 0.75–1 ml solvent). The

solvent was removed using a vacuum oven setup inside the glovebox where the samples were initially kept at ambient temperature while the pressure was gradually decreased from an initial 200 mbar to <1 mbar during 20 h, followed by heating at 60 °C and <1 mbar for 40 h. After cooling to room temperature, circular samples with a diameter of 14 mm were punched from the resulting films. Great care was taken to avoid moisture contamination, including conducting all monomer, polymer and electrolyte handling in a glovebox under argon. The polymer is obtained anhydrous from synthesis, as any trace amounts of water in the reaction mixture will be consumed during polymerization.

2.3. Thermal properties

The thermal properties of copolymers and electrolytes were determined through differential scanning calorimetry on a TA Instruments DSC Q2000. Samples were hermetically sealed in aluminum pans under argon. The pans were rapidly cooled to –80 °C and subsequently heated at a rate of 10 °C/min to 130 °C for measurement. This cycle was repeated to assess the tendency for recrystallization of the polymers.

2.4. Ionic conductivity

The ionic conductivity of the electrolyte films was measured using electrochemical impedance spectroscopy. Samples were sandwiched between stainless steel blocking electrodes and sealed in Swagelok-type cells. The thickness of the electrolyte films was measured prior to cell assembly using a Mitutoyo digital indicator. The measurement cells were designed to exert a minimum amount of pressure on the electrolyte film to ensure proper electrical contact while avoiding deformation of the sample. The cells were heated to 100 °C in an oven while monitoring the cell temperature using a thermocouple inserted a few mm from one of the electrodes. The cells were kept at 100 °C for 30 min to ensure good interfacial contacts and were then allowed to cool and left at room temperature for 20 h to allow for recrystallization. The ionic conductivity was measured during heating using an SI 1260 Impedance/Gain-Phase Analyzer (Schlumberger) at a frequency range of 1 Hz–10 MHz and an amplitude of 10 mV. The ionic conductivity was obtained from the Nyquist plots as the (low-frequency) intercept with the real axis.

2.5. Transference number

Lithium transference numbers were measured using potentiostatic polarization [21]. Symmetric Li/SPE/Li cells were assembled and thermally equilibrated for 20 h (60 °C measurement) or 48 h (40 °C measurements) and the impedance in the cell was monitored until a stable SEI was observed (no obvious change in resistance). Impedance and polarization measurements were carried out using a VMP2 (Biologic). The cells were polarized at 10 mV and the impedance spectra were obtained from 1 Hz to 200 kHz (40 °C measurements) or 1 Hz to 1.5 MHz (60 °C measurement) at an amplitude of 10 mV. Temperature fluctuations were maintained within 0.1 °C.

2.6. Battery cycling

The preparation of LiFePO₄ cathodes and Li half-cell assembly has already been described elsewhere [14]. The full capacity of the active material, as reported by the manufacturer, is approximately 150 mAh/g. The active mass loadings of the electrodes was in the range of 1.8–2.5 mg/cm² and the thickness around 13–16 μ m. Electrolytes for Li half-cell assembly were solution-cast directly

onto LiFePO_4 cathodes using the same evaporation procedure as described above for the preparation of electrolyte films. Before cycling at 40 °C, the assembled cells were stored at 40 °C for 3 h. Cells were cycled between 2.7 and 4.2 V using a Digatron BTS-600.

3. Results and discussion

The issue of crystallinity in the PCL system can be addressed by incorporating small amounts of TMC as a comonomer at synthesis [22], thereby disrupting the regularity of the structure and suppressing the crystallinity of the material. Indeed, electrolytes prepared from high-molecular-weight (M_n in the range 246 000–338 000 g/mol, see Table S1 in the Supplementary material) random copolymers of ϵ -caprolactone (CL) and TMC doped with lithium bis(trifluoromethanesulfonimide) (LiTFSI) salt are largely amorphous and show sufficient mechanical stability without the need for crosslinking, thus allowing for easy handling and processing. As anticipated, these SPEs display substantially increased ionic conductivity at temperatures below the melting point of PCL crystallites (Fig. 1). More remarkably, however, it was found that adding up to 20 mol% of the carbonate repeating unit to the host polymer further increases the ionic conductivity even at temperatures exceeding the semicrystalline region of PCL. In flexible, amorphous polymer electrolytes, ion transport is typically dependent on the segmental motions of the host polymer [23]. Given the lower molecular flexibility of the TMC repeating unit, as indicated by the higher glass transition temperature of PTMC

compared to that of PCL (−16 °C and −63.7 °C, respectively), an increased TMC content would be expected to lower the ionic conductivity. Instead, the highest conductivity can be seen for electrolytes prepared from the copolymer with a molar composition of 80:20 CL:TMC (Fig. 2). Electrolytes prepared from this copolymer with ≥ 28 wt% LiTFSI are fully amorphous, as indicated by DSC measurements, and show markedly improved ionic conductivities compared to electrolytes prepared from homopolymers of either substituent monomer [14].

As can be seen in Fig. 1, comparatively impressive conductivities were obtained for the 80:20 copolymer at a salt content of 28–36 wt%, with the latter electrolyte showing slightly higher ionic conductivity throughout the investigated temperature range, as is evident from Fig. 1b. With a total ionic conductivity of $4.1 \times 10^{-5} \text{ S cm}^{-1}$ at 25 °C, the performance of this electrolyte can be favorably compared to even the best-performing polyether systems [6] and is close to what has been suggested as the upper limit for polyether electrolytes [12].

While necessary for fast ion transport, a high total ionic conductivity does not in itself guarantee efficient battery operation. Specifically, a lithium transference number (t_+) close to unity is also desired. Polyether electrolytes typically suffer from low lithium transference numbers, thus leaving the anions responsible for the bulk of the ion transport and rendering the effective lithium conductivity low [12,24]. In contrast, as can be seen in Fig. 3, the 80:20 electrolyte with 36 wt% LiTFSI (henceforth denoted as 80:20₃₆) displays high lithium transference numbers of 0.66 and 0.62 at 60

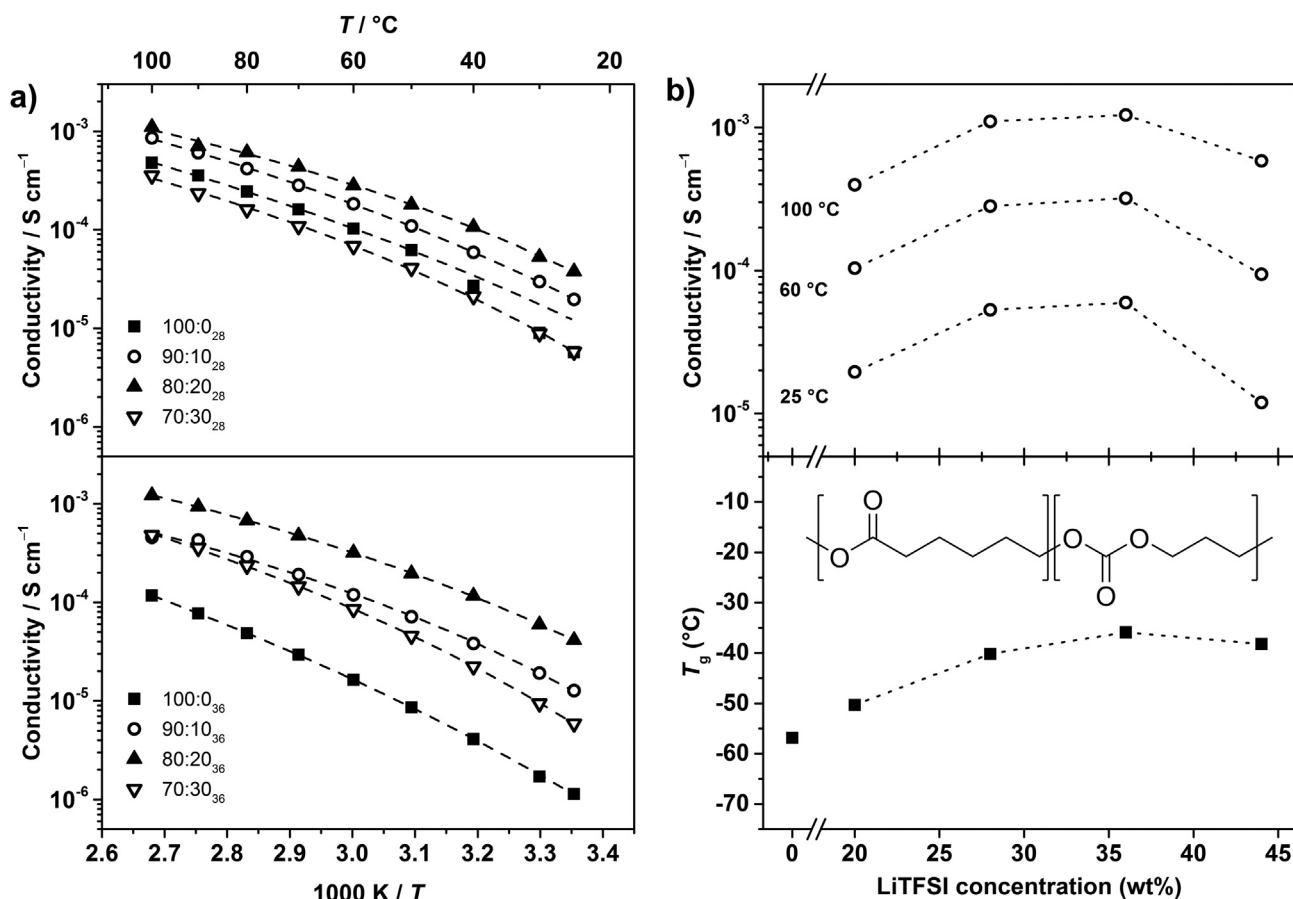


Fig. 1. a) Conductivity comparison for polymer electrolytes with 28 wt% LiTFSI (top) and 36 wt% LiTFSI (bottom). The electrolytes are identified by the CL:TMC ratios with the LiTFSI concentrations (wt%) as subscripts. Dashed lines represent VTF fits. For the semicrystalline 100:0₂₈ electrolyte, the VTF fit at high temperatures has been extrapolated below the melting point to illustrate the effect of crystallization. b) Conductivity isotherms at 25, 60 and 100 °C (top) and glass transition temperature (bottom) for PCL–PTMC 80:20 copolymer electrolytes. The inset shows the structure of the PCL–PTMC copolymers.

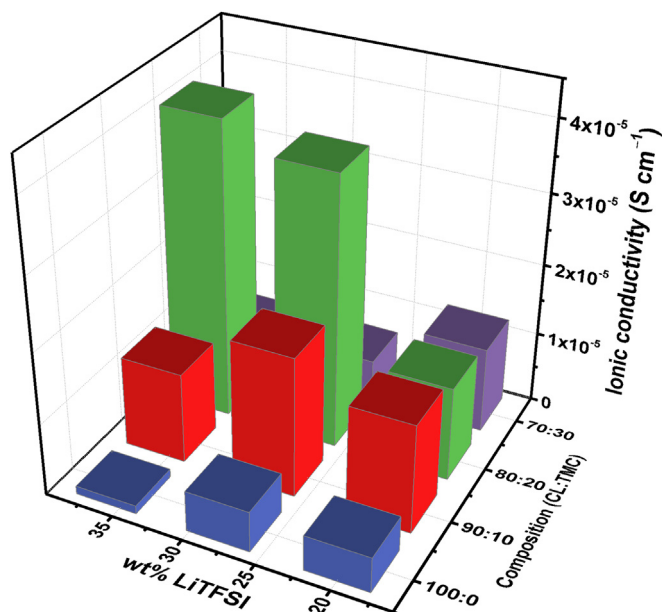


Fig. 2. Total ionic conductivities for the poly(CL-co-TMC)–LiTFSI electrolyte system at 25 °C.

and 40 °C, respectively, as determined through the potentiostatic polarization method [21]. The close correspondence between these measurements at elevated temperatures indicates a high t_+ in a broad temperature range, which should also translate to a high Li^+

transference number at room temperature, although this was not possible to confirm using the current measurement setup. For comparison, a reference PEO electrolyte comprising 28 wt% LiTFSI shows a t_+ of 0.12, i.e., far below that of the PCL–PTMC copolymer electrolyte and in line with what has been previously reported for other polyether electrolytes [15,24]. For the PEO sample, the effects of crystallinity can be clearly seen in the impedance plots as a notably high interfacial resistance.

The clearly improved Li^+ transport properties on addition of the carbonate repeating units to the PCL polyester backbone indicate that the presence of carbonate groups results in overall weaker binding between the coordinating carbonyl groups and the Li^+ ions. This could be either due to inherently weaker coordination between the carbonate groups and Li^+ or due to disruption of the sequence regularity of the PCL backbone, thereby preventing chelating effects that lead to stronger binding and impeded Li^+ transport.

The combination of high room-temperature ionic conductivity and lithium transference number of the 80:20₃₆ electrolyte prompted us to test the performance of this new material in LiFePO_4 half-cells operated at ambient temperature. Room-temperature all-solid lithium polymer batteries have largely been dismissed as infeasible and battery cells with non-plasticized polymer electrolytes are typically cycled at 50–80 °C in order to achieve reasonable performance [25–28]. Even with plasticizing and/or conductivity-enhancing additives, it is not trivial to attain reasonable performance at temperatures below this range [29,30] and it has been demonstrated that battery cells with traditional PEO-based electrolytes without additives can hardly be cycled at room temperature with more than very limited capacity [31]. Consequentially, examples of lithium batteries with solid polymer electrolytes without low-molecular-weight additives operating at lower temperatures are few and far between. Strategies employed to obtain appropriate ion transport properties of the electrolyte include PEO derivatives rendered amorphous by copolymerization [28,32,33], blends of PEO with a PEO-based macromonomer [34], copolymers with flexible PEO- and polysiloxane-based side chains [35] and polysiloxane-based copolymers with grafted PEO chains [36]. Commonly, these strategies are equivalent to significant compromises with the mechanical integrity of the SPE. Examples include low-molecular-weight polymers based on polysiloxane backbones without inherent mechanical stability that require stabilization with poly(vinylidene fluoride) in order to function as a solid separator in working battery cells [37,38].

In contrast, the PCL–PTMC copolymer electrolytes have sufficient mechanical stability to function as an interelectrode separator and, as is evident from the data in Fig. 4, lithium half-cells assembled using the 80:20₃₆ electrolyte can be cycled at ambient temperature (22.4–22.8 °C), reaching >80% of the full capacity (150 mA h/g) at C/50 and maintaining useful capacities at up to C/10 (corresponding to a current density of 0.027 mA cm^{−2}). Most importantly, the capacity retention was found to be excellent with no discernible loss in capacity over >70 cycles (see Figure S3, Supplementary material) as well as coulombic efficiencies close to 100%, much in line with our previous findings on similar amorphous polymer electrolyte systems based on polyesters and polycarbonates [14,17]. As shown in Fig. 4, increasing the temperature slightly to a modest 40 °C improved the capacity further for the 80:20₃₆-based cells with unimpaired capacity retention and coulombic efficiencies at rates up to C/5 (Figure S4, Supplementary material).

4. Conclusions

In conclusion, we have demonstrated that, while the conductivity of poly(trimethylene carbonate) electrolytes can be enhanced

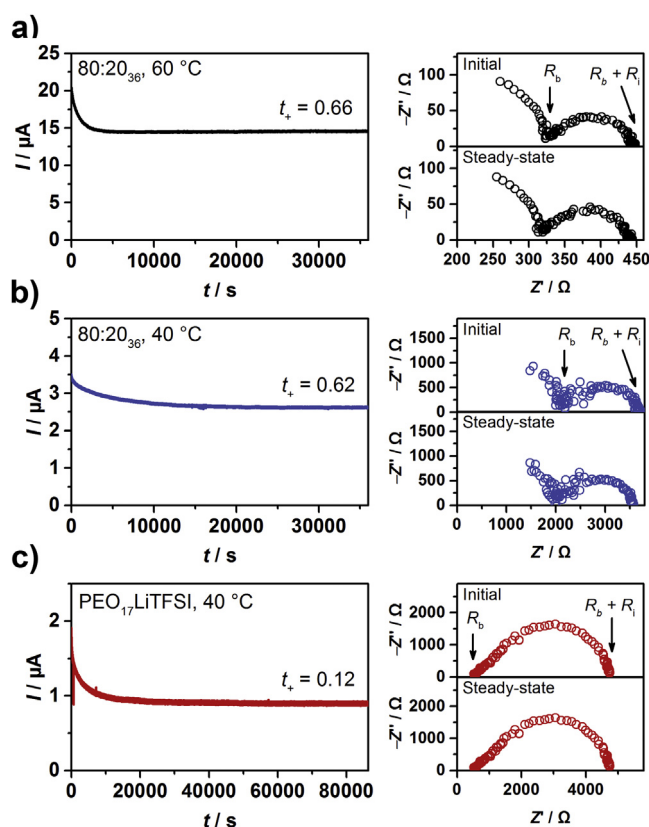


Fig. 3. Polarization curves (left), and initial and steady-state impedance plots (right) for determination of lithium transference numbers (t_+) by potentiostatic polarization of symmetric $\text{Li}/\text{SPE}/\text{Li}$ cells at 10 mV. R_b refers to the bulk ionic resistance and R_i refers to the interfacial resistance. a) PCL–PTMC 80:20 with 36 wt% LiTFSI, 60 °C; b) PCL–PTMC 80:20 with 36 wt% LiTFSI, 40 °C; c) $\text{PEO}_{17}\text{LiTFSI}$ ([ether]:[Li^+] = 17), 40 °C.

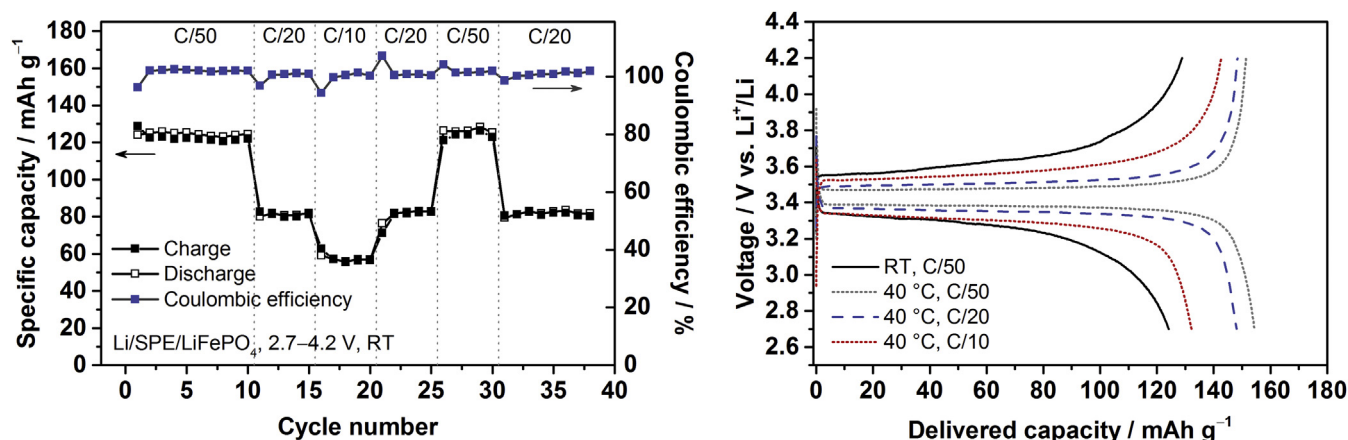


Fig. 4. Charge/discharge capacities and coulombic efficiencies for a Li/SPE/LiFePO₄ cell with the 80:20₃₆ electrolyte, cycled at room temperature (left) and initial charge/discharge voltage curves comparing Li/SPE/LiFePO₄ cells with the 80:20₃₆ electrolyte cycled at different temperatures and C-rates (right).

by the addition of plasticizing ϵ -caprolactone repeating units, similar improvements in ionic conductivity can be attained also for the opposite end of the composition spectrum, i.e., by adding trimethylene carbonate repeating units to a poly(ϵ -caprolactone) backbone. The inclusion of the carbonate repeating units in the host polymer results in electrolytes that are largely amorphous, mechanically stable and, most importantly, display markedly increased ionic conductivities rivaling those of the very best poly-ether electrolytes, making them suitable for use in rechargeable all-solid Li polymer batteries that can be operated even at ambient temperature. The stable performance demonstrated at room temperature should make such cells suitable for low-drain applications where stability is essential, while a modest increase in temperature markedly improves the performance also at higher discharge rates. The facile synthesis and favorable Li⁺ transport properties of these electrolytes make this a highly promising electrolyte system. We anticipate that with further improvements, it will be possible to extend the performance of these all-solid Li polymer cells to also encompass high discharge rates at ambient temperature in order to realize the full potential of solid polymer electrolyte batteries.

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Appendix A. Supplementary material

Supplementary material related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2015.08.035>.

References

- [1] J.B. Goodenough, Y. Kim, *Chem. Mater.* 22 (2010) 587–603.
- [2] R. Fong, U. Vonsacken, J.R. Dahn, *J. Electrochem. Soc.* 137 (1990) 2009–2013.
- [3] K. Xu, *Chem. Rev.* 104 (2004) 4303–4418.
- [4] E.P. Roth, C.J. Orendorff, *Electrochem. Soc. Interface* 21 (2012) 37–41.
- [5] N. Williard, W. He, C. Hendricks, M. Pecht, *Energies* 6 (2013) 4682–4695.
- [6] J.W. Fergus, *J. Power Sources* 195 (2010) 4554–4569.
- [7] K. Edström, D. Brandell, T. Gustafsson, L. Nyholm, *Electrochem. Soc. Interface* 20 (2011) 41–46.
- [8] M. Armand, J.M. Chabagno, M. Duclot, in: *Second International Meeting on*

- Solid Electrolytes*, St. Andrews, Scotland, 1978.
- [9] M.B. Armand, J.M. Chabagno, M.J. Duclot, Poly-ethers as solid electrolytes, in: P. Vashishta, J.N. Mundy, G.K. Shenoy (Eds.), *Fast Ion Transport in Solids: Electrodes and Electrolytes*, Elsevier North Holland, New York, 1979, pp. 131–136.
- [10] B. Sun, I.Y. Liao, S. Tan, T. Bowden, D. Brandell, *J. Power Sources* 238 (2013) 435–441.
- [11] J.F. Snyder, R.H. Carter, E.D. Wetzel, *Chem. Mater.* 19 (2007) 3793–3801.
- [12] O. Buriez, Y.B. Han, J. Hou, J.B. Kerr, J. Qiao, S.E. Sloop, M. Tian, S. Wang, *J. Power Sources* 89 (2000) 149–155.
- [13] K.P. Barteau, M. Wolffs, N.A. Lynd, G.H. Fredrickson, E.J. Kramer, C.J. Hawker, *Macromolecules* 46 (2013) 8988–8994.
- [14] B. Sun, J. Mindemark, K. Edström, D. Brandell, *Solid State Ionics* 262 (2014) 738–742.
- [15] Y. Tominaga, K. Yamazaki, *Chem. Commun.* 50 (2014) 4448–4450.
- [16] B. Sun, J. Mindemark, K. Edström, D. Brandell, *Electrochem. Commun.* 52 (2015) 71–74.
- [17] J. Mindemark, E. Törmä, B. Sun, D. Brandell, *Polymer* 63 (2015) 91–98.
- [18] C.P. Fonseca, S. Neves, *J. Power Sources* 159 (2006) 712–716.
- [19] C.P. Fonseca, D.S. Rosa, F. Gaboardi, S. Neves, *J. Power Sources* 155 (2006) 381–384.
- [20] C.K. Lin, I.D. Wu, *Polymer* 52 (2011) 4106–4113.
- [21] J. Evans, C.A. Vincent, P.G. Bruce, *Polymer* 28 (1987) 2324–2328.
- [22] A.P. Pego, A.A. Poot, D.W. Grijpma, J. Feijen, *J. Biomater. Sci. Polym. Ed.* 12 (2001) 35–53.
- [23] Y. Wang, F. Fan, A.L. Agapov, T. Saito, J. Yang, X. Yu, K. Hong, J. Mays, A.P. Sokolov, *Polymer* 55 (2014) 4067–4076.
- [24] D. Devaux, R. Bouchet, D. Glé, R. Denoyel, *Solid State Ionics* 227 (2012) 119–127.
- [25] Y. Kang, H.J. Kim, E. Kim, B. Oh, J.H. Cho, *J. Power Sources* 92 (2001) 255–259.
- [26] Y. Kobayashi, H. Miyashiro, K. Takei, H. Shigemura, M. Tabuchi, H. Kageyama, T. Iwahori, *J. Electrochem. Soc.* 150 (2003) A1577–A1582.
- [27] Y. Aihara, J. Kuratomi, T. Bando, T. Iguchi, H. Yoshida, T. Ono, K. Kuwana, *J. Power Sources* 114 (2003) 96–104.
- [28] M. Gauthier, D. Fauteux, G. Vassort, A. Belanger, M. Duval, P. Ricoux, J.M. Chabagno, D. Muller, P. Rigaud, M.B. Armand, D. Deroo, *J. Power Sources* 14 (1985) 23–26.
- [29] J.H. Shin, W.A. Henderson, S. Scaccia, P.P. Prosini, S. Passerini, *J. Power Sources* 156 (2006) 560–566.
- [30] M. Wetjen, G.-T. Kim, M. Joost, M. Winter, S. Passerini, *Electrochim. Acta* 87 (2013) 779–787.
- [31] G.B. Appetecchi, G.T. Kim, M. Montanino, F. Alessandrini, S. Passerini, *J. Power Sources* 196 (2011) 6703–6709.
- [32] M. Gauthier, D. Fauteux, G. Vassort, A. Belanger, M. Duval, P. Ricoux, J.M. Chabagno, D. Muller, P. Rigaud, M.B. Armand, D. Deroo, *J. Electrochem. Soc.* 132 (1985) 1333–1340.
- [33] M. Armand, D. Muller, M. Duval, P. E. Harvey, Societe Nationale Elf Aquitaine, Hydro-Quebec. Novel Macromolecular Material for use in Realizing Electrolytes and/or Electrodes. US patent 4,578,326. 1986 Mar 25.
- [34] C. Arbizzani, M. Mastragostino, T. Hamaide, A. Guyot, *Electrochim. Acta* 35 (1990) 1781–1785.
- [35] P.E. Trapa, Y.Y. Won, S.C. Mui, E.A. Olivetti, B.Y. Huang, D.R. Sadoway, A.M. Mayes, S. Dallek, *J. Electrochem. Soc.* 152 (2005) A1–A5.
- [36] F.-M. Wang, C.-C. Hu, S.-C. Lo, Y.-Y. Wang, C.-C. Wan, *Solid State Ionics* 180 (2009) 405–411.
- [37] Y. Lin, J. Li, Y. Lai, C. Yuan, Y. Cheng, J. Liu, *RSC Adv.* 3 (2013) 10722–10730.
- [38] J. Li, Y. Lin, H. Yao, C. Yuan, J. Liu, *ChemSusChem* 7 (2014) 1901–1908.